

Terrain-Conditioned Cross-Attention for Cross-Region Avalanche Debris Mapping with Sentinel-1 SAR

1

¹Author affiliations withheld for double-blind peer review.

Abstract

Cloud cover, winter darkness, and steep terrain complicate regional mapping of avalanche debris. Sentinel-1 synthetic aperture radar (SAR) provides all-weather observations, but slope-dependent foreshortening, layover, and shadow can resemble bitemporal change. We present TopoRadar-Net, a lightweight change-segmentation model that conditions SAR features on local incidence angle, slope, aspect, elevation, and a fixed-reference aspect coordinate through multiscale cross-attention. We evaluate five architectures on AvalCD, comprising seven events, four mountain systems, and 993 avalanche polygons. Models are trained on Alps and Greenland scenes and tested without site-specific tuning in Tromsø and the Pamirs. Across three seeds, TopoRadar-Net obtains **63.79% $\pm$ 1.88%** pooled macro F1 (mean $\pm$ population standard deviation). Its margin over a bottleneck self-attention U-Net is **+0.38** percentage points ($p = \mathbf{0.869}$), whereas its margin over SiamUNet-diff is **+8.92** percentage points ($p = \mathbf{0.021}$). Ablations show regional rather than uniform benefits: removing aspect conditioning or cross-attention increases pooled F1, while the slope prior overlaps **18.47%** of training positives. A validation-calibrated road-corridor proxy reaches **83.0%** precision and **62.5%** recall in the Pamirs but misses the single Tromsø corridor avalanche in every seed. Terrain conditioning is therefore competitive and diagnostically useful, but not uniformly superior or deployment-ready.

Keywords: Sentinel-1 SAR, snow avalanche mapping, semantic segmentation, radar geometry, cross-region evaluation, deep learning

047 1 Introduction

048
049 Regional avalanche-debris maps support event documentation, hazard assessment,
050 and prioritization of field inspection. Optical observations are often unavailable during
051 major winter storms, while Sentinel-1 C-band synthetic aperture radar (SAR) supplies
052 day-and-night measurements through cloud cover (Eckerstorfer, Malnes, & Müller,
053 2017; Eckerstorfer, Müller, Malnes, & Oterhals, 2021; Eckerstorfer, Vickers, Malnes,
054 & Grahn, 2019). Avalanche deposits roughen the snow surface and can increase
055 post-event co- and cross-polarized backscatter, making bitemporal change detection
056 useful (Tompkin & Leinss, 2021). AvalCD supplies VV/VH for Livigno, Tromsø, and
057 Pish, but HH/HV for Nuuk. Side-looking geometry also produces slope-dependent
058 foreshortening, layover, shadow, and radiometric variation that can resemble debris.
059

060 Recent remote-sensing studies show that deep learning can map slope hazards
061 across diverse environments, but shadows, scale changes, class imbalance, and geo-
062 graphic transfer remain difficult (Chandra, Vaidya, Chauhan, & Shaw, 2025; Nguyen
063 et al., 2025). SAR-specific segmentation and change-image construction introduce
064 additional representation choices (Arjasakusuma & Kusuma, 2025; Wang et al., 2025).
065 In the same hazard domain, Ahmad, Mallick, Hang, Rihan, and Bakshi (2026) model
066 pre-event avalanche susceptibility with attention, whereas our target is post-event
067 debris. Debris mapping for emergency response also needs explicit comparison and
068 operationally meaningful validation rather than accuracy alone (Bayat, Dindar, &
069 Erkal, 2026).

070 The closest avalanche-segmentation precedents already contain terrain condition-
071 ing. Bianchi, Grahn, Eckerstorfer, Malnes, and Vickers (2021) apply a potential-angle-
072 of-reach-conditioned attention mask and slope feature map to Sentinel-1 avalanche
073 segmentation. On AvalCD, Gatti, Mariani, Gallo, and Monti (2026) encode local inci-
074 dence angle and slope in a separate transformer branch and report that SAR-only
075 change detection is more consistent than terrain-augmented variants. Concurrently,
076
077
078
079
080
081
082
083
084
085
086
087
088
089
090
091
092

Gelato, Sgaravatti, Grahm, Boracchi, and Bianchi (2026) adapt a promptable foundation model to SAR avalanche segmentation using SAR, elevation, and slope inputs and evaluate annotation-time reduction. Our target instead is autonomous bitemporal change segmentation across held-out regions, with multi-seed failure characterization rather than prompt-driven annotation. Our architectural delta is narrower: SAR-change queries attend to pooled terrain keys and values, including a fixed 78° aspect-reference channel. The empirical contribution is the three-seed, leave-region-out characterization of component instability, positive-label conflict in the slope prior, and a bounded corridor proxy—not a claim that terrain conditioning uniformly improves accuracy.

We evaluate whether radar-topographic conditioning improves avalanche change segmentation across unseen mountain systems. TopoRadar-Net uses bitemporal SAR differences as queries and multiscale terrain geometry as keys and values. The study contributes: (1) a compact terrain-conditioned cross-attention architecture; (2) three-seed zero-shot evaluation against four baselines on two held-out mountain systems; (3) spatial-block inference, component ablations, and terrain-stratum stability checks; and (4) construct and corridor audits that expose where the model should not be interpreted operationally. The objective is not to claim uniform superiority, but to establish where terrain conditioning changes performance and where its assumptions fail.

2 Data and Study Design

AvalCD contains co-registered pre/post-event Sentinel-1 intensity, topography, valid-data masks, and expert-mapped avalanche polygons (Gatti, 2026; Gatti et al., 2026). The seven events span the European Alps, Greenland Arctic, Scandinavian Arctic, and Pamir Mountains (Table 1). Livigno events from April 2024 and January 2025 plus Nuuk 2016 form the training cohort. Livigno March 2025 and Nuuk 2021 are

Table 1 AvalCD scenes used in the cross-region study. Debris is the number of labelled positive pixels

System	Scene date	Role	Region	Pol.	Polygons	Debris
Alps	3 Apr 2024	Train	Livigno	VV/VH	231	32,606
Alps	29 Jan 2025	Train	Livigno	VV/VH	207	11,438
Alps	18 Mar 2025	Validate	Livigno	VV/VH	79	7,392
Greenland	13 Apr 2016	Train	Nuuk	HH/HV	117	64,822
Greenland	11 Apr 2021	Validate	Nuuk	HH/HV	129	81,595
Pamirs	21 Feb 2023	Test	Pish	VV/VH	113	22,579
Scand. Arctic	20 Dec 2024	Test	Tromsø	VV/VH	117	33,972

used for validation, checkpoint selection, and decision-threshold calibration. Tromsø 2024 and Pish 2023 remain geographically held out for final testing. This split changes climate, relief, polarization regime, acquisition geometry, and avalanche morphology simultaneously; results quantify cross-system transfer but do not identify any one changed factor as causal.

The executed loader uses the supplied decibel rasters without percentile scaling. Values outside $[-45, 15]$ dB are marked invalid; after the valid mask is fixed, missing cross- and co-polarized values are replaced by -20 and -12 dB, respectively. The two model positions represent cross/co polarization (VH/VV or HV/HH according to scene), and difference channels retain those positions. The terrain tensor contains local incidence angle (LIA), slope, sine/cosine aspect, elevation from a digital elevation model (DEM), and the fixed coordinate $\cos(\alpha - 78^\circ)$; this last channel is a reference-axis aspect encoding, not authoritative per-event radar heading. Training uses balanced 128×128 patches; evaluation uses complete valid-mask rasters. Physical areas are derived from affine raster transforms rather than full rectangular extents. Figure 1 summarizes the model; Online Resource 1 supplies equations, preprocessing details, complete tables, and additional diagnostics.

3 Method

3.1 Radar-topographic conditioning

Let I^{pre} and I^{post} denote pre- and post-event two-channel cross/co-polarized intensity tensors. A shared residual encoder produces scale- l features F_l^{pre} and F_l^{post} . Their difference representation is

$$F_l^\Delta = \text{Conv}[F_l^{\text{post}} - F_l^{\text{pre}}, F_l^{\text{post}}, F_l^{\text{pre}}]. \quad (1)$$

The terrain encoder maps the six-channel geometry tensor to T_l . A Radar-Topographic Cross-Attention Block (RTCAB) forms SAR queries and pooled terrain keys and values,

$$Q = F_l^\Delta W_Q, \quad K = \text{Pool}(T_l) W_K, \quad V = \text{Pool}(T_l) W_V, \quad (2)$$

and computes $A = \text{softmax}(QK^\top / \sqrt{d})V$. A pixelwise gate $\sigma(\text{Conv}(T_l))$ modulates the attended features before progressive decoding. This construction conditions a shift-invariant convolutional representation on spatially varying radar geometry; it does not make the representation physically invariant.

3.2 Loss, training, and inference

The objective combines positive-weighted binary cross-entropy (BCE), Dice loss, and a soft slope prior:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + 0.5 \mathcal{L}_{\text{slope}}. \quad (3)$$

The prior discourages probability mass below 5° and above 65° but is treated as a heuristic because mapped positives occur in this mask. All models use the same optimizer, six-epoch schedule, balanced patches, validation-selected checkpoint, and full-scene threshold calibration. We train fixed seeds 42, 123, and 456. Comparators

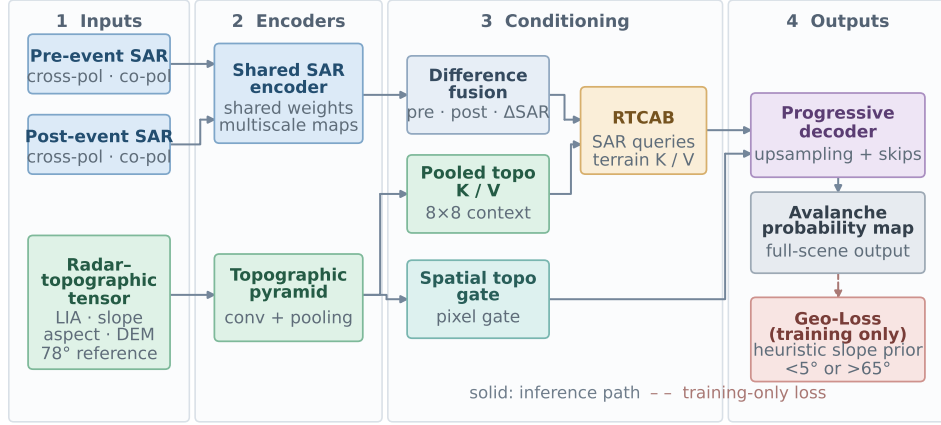


Fig. 1 TopoRadar-Net schematic. Bitemporal SAR differences query pooled topographic keys and values; a pixelwise terrain gate conditions decoding. The training schematic omits the ground-truth loss input for clarity

are SiamUNet-diff and SiamUNet-conc (Caye Daudt, Le Saux, & Boulch, 2018), ResU-Net (Zhang, Liu, & Wang, 2018), and an internal bottleneck self-attention U-Net. The internal baseline uses the same convolutional encoder-decoder scale with four-head self-attention only at the deepest feature map; it is neither canonical attention-gated U-Net nor Swin-UNet. Parameter counts range from 2.01 to 3.46 million.

Inference uses overlapping windows with Hanning blending. Pixel F1 and intersection-over-union are computed over valid pixels. Polygon hit rate requires at least 30% overlap. A 1,000-draw paired bootstrap resamples 128×128 spatial blocks; paired seed-level tests use the three macro-F1 values. The latter have low power and are interpreted together with effect sizes and ordering across seeds. No test pixel or test label is used for model or threshold selection.

For a post-hoc operational proxy, validation-scene component scores calibrate a one-feature logistic model and an F2-optimal alert threshold. Frozen test-scene alerts are intersected with 30-m buffers around a versioned OpenStreetMap major-road snapshot (OpenStreetMap contributors, 2026). This retrospective proxy excludes data

Table 2 Zero-shot test performance across three fixed training seeds (mean $\pm$ descriptive population standard deviation). HR is polygon hit rate at 30% overlap

Method	Tromsø F1	Pish F1	Pooled F1	Tromsø HR
SiamUNet-diff	72.64 $\pm$ 0.39	37.11 $\pm$ 2.44	54.87 $\pm$ 1.03	63.25 $\pm$ 2.52
SiamUNet-conc	78.92 $\pm$ 1.07	47.34 $\pm$ 0.70	63.13 $\pm$ 0.79	82.62 $\pm$ 1.07
ResU-Net	76.60 $\pm$ 1.30	49.25 $\pm$ 3.39	62.92 $\pm$ 1.42	76.92 $\pm$ 4.83
Self-attention U-Net	78.09 $\pm$ 0.99	48.73 $\pm$ 2.74	63.41 $\pm$ 1.03	79.20 $\pm$ 2.45
TopoRadar-Net	77.40 $\pm$ 1.86	50.18 $\pm$ 4.40	63.79 $\pm$ 1.88	75.50 $\pm$ 1.61

acquisition, human review, dispatch, and closure decisions. Figures 1 and 2 were generated with Python 3.14.7 and Matplotlib 3.11.2.

3.3 AI-assisted research support

Generative AI tools assisted code refactoring, manuscript language, and checklist-based consistency checking. They did not generate research data or determine final scientific claims. The authors inspected generated material, reran the analysis and verification code, checked citations and numerical statements, and retain full responsibility for the work.

4 Results

4.1 Cross-region performance

TopoRadar-Net records 77.40% $\pm$ 1.86% F1 in Tromsø and 50.18% $\pm$ 4.40% in Pish, giving 63.79% $\pm$ 1.88% pooled macro F1 (Table 2). The bottleneck self-attention U-Net obtains 63.41% $\pm$ 1.03%, SiamUNet-conc 63.13% $\pm$ 0.79%, and ResU-Net 62.92% $\pm$ 1.42%. Their differences from TopoRadar-Net are not significant across three seeds. SiamUNet-diff is lower at 54.87% $\pm$ 1.03%; the paired difference is +8.92 percentage points (95% confidence interval [CI] +3.25 to +14.59, $p = 0.021$). The bottleneck self-attention U-Net leads on seed 123, demonstrating initialization-dependent ordering. Figure 2 visualizes the region-specific ordering.

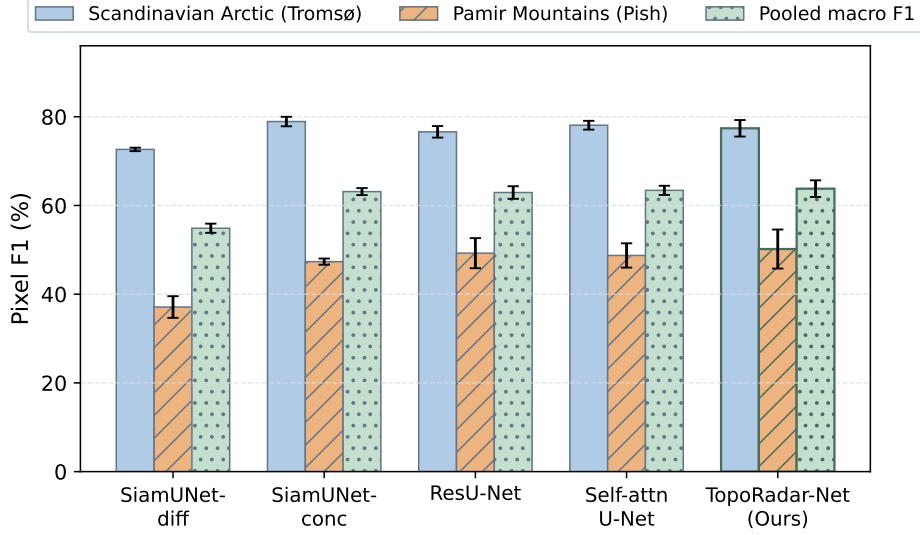


Fig. 2 Zero-shot pixel F1 across held-out Tromsø, Pish, and their macro average. Error bars show descriptive population standard deviation over three fixed training seeds

The 339-block seed-42 bootstrap gives a +3.27-percentage-point micro-F1 difference from the self-attention U-Net (95% CI +1.12 to +5.37, $p = 0.010$), but repeating the bootstrap by training seed yields +3.26, -2.99, and +4.16 percentage points. Thus a conditional spatial interval is not a substitute for training uncertainty. On the Nuuk validation scene, all modern architectures lie within 0.63 F1 percentage points, providing additional evidence of parity rather than uniform dominance.

Online Resource 1 reports polygon completeness by European Avalanche Warning Services destructive-size classes ([European Avalanche Warning Services, 2022](#)); these classes are available for Tromsø but not Pish.

4.2 Component sensitivity and failure boundaries

The full architecture is not the best pooled configuration (Table 3). Removing aspect and its fixed-reference coordinate or removing cross-attention increases pooled macro F1 to 65.35% and 65.07%, respectively, mainly through gains in Pish. In Tromsø, seed-42 paired block tests favour the full model over No-LIA by 2.04 percentage points ($p <$

Table 3 Ablation performance across three seeds. The highest pooled value is obtained without aspect and its fixed-reference coordinate

Configuration	Tromsø F1	Pish F1	Pooled F1
Full	77.40 ± 1.86	50.18 ± 4.40	63.79 ± 1.88
No local incidence angle	75.67 ± 1.76	52.13 ± 1.11	63.90 ± 1.37
No cross-attention	76.85 ± 0.80	53.30 ± 0.93	65.07 ± 0.53
No slope prior	76.70 ± 0.91	51.99 ± 1.31	64.35 ± 1.05
No aspect/reference coordinate	77.38 ± 2.66	53.32 ± 0.42	65.35 ± 1.54

0.001) and No-CrossAttn by 1.42 percentage points ($p = 0.010$). These comparisons establish component sensitivity, not a causal terrain-correction mechanism.

The slope-prior mask contains 18.47% of valid training positives, 0.95% of validation positives, and 0.87% of held-out positives. Within the held-out penalty stratum, the full model has lower three-seed positive recall than No-GeoLoss in Tromsø ($50.47\% \pm 3.18\%$ versus $71.29\% \pm 6.39\%$) and Pish ($1.08\% \pm 1.52\%$ versus $6.45\% \pm 6.97\%$). The term is therefore a precision–recall prior, not a physical-admissibility rule.

Seed-42 terrain strata initially suggest positive backslope differences, but these do not persist across seeds: TopoRadar-Net minus the self-attention U-Net averages $-0.34\% \pm 6.28\%$ in Tromsø and $+1.20\% \pm 7.73\%$ in Pish. Extreme-steep terrain is highly variable and both models perform poorly in the small Pish shadow stratum. Complete stratum results are in Online Resource 1.

4.3 Retrospective corridor proxy

The affine valid-mask area is 196.820 km^2 in Tromsø and 267.435 km^2 in Pish. In Tromsø, TopoRadar-Net produces 74.77 ha of false positives at 0.38 ha km^{-2} . In Pish, it produces 129.69 ha at 0.48 ha km^{-2} . Relative to the self-attention U-Net, this is a 30.0% reduction in Pish (129.69 versus 185.32 ha) but a 28.5% increase in Tromsø (74.77 versus 58.19 ha). Validation-calibrated road-corridor alerts achieve 83.0% precision and 62.5% recall in Pish. The single road-intersecting reference avalanche in

415 Tromsø is missed by every TopoRadar-Net seed. All cached complete-scene runs
416 are below 5 s on Apple Metal Performance Shaders hardware, but timing excludes
417 upstream acquisition and human review.
418
419

420

421 5 Discussion

422

423
424 TopoRadar-Net is competitive across unseen mountain systems, but the strongest
425 baselines are statistically indistinguishable across three seeds. This result is impor-
426 tant for remote-sensing model design: adding physically interpretable variables does
427 not guarantee a better predictor. Recent hazard-mapping studies similarly motivate
428 multiscale or attention modules across diverse environments ([Chandra et al., 2025](#);
429 [Nguyen et al., 2025](#)), while our cross-region ablations show that auxiliary geometry
430 can reverse model ordering. The contribution is therefore the controlled character-
431 ization of terrain conditioning under geographic shift, including evidence against a
432 uniform advantage.
433

434 The construct audit changes the interpretation of the slope loss. Genuine positives
435 inside the penalty mask arise through runout, raster alignment, and terrain-model
436 uncertainty; suppressing them can increase precision while reducing recall. A hard
437 “impossible terrain” label would be unsupported. Likewise, seed-42 terrain-stratum
438 differences are descriptive because seed repetition does not preserve their magnitude
439 or sign.

440 Operationally, the component and corridor analyses resemble recent efforts to
441 connect mapped debris with transport access ([Bayat et al., 2026](#)), but they remain
442 retrospective. The Pish corridor result is encouraging; the Tromsø miss falsifies any
443 claim of region-independent completeness. Prospective evaluation with warning ser-
444 vices or road authorities must measure acquisition delay, review burden, missed-event
445 cost, and decision outcomes before deployment.
446
447
448

449

450

451

452

453

454

455

456

457

458

459

460

Limitations include seven events from one benchmark, three training seeds, correlated spatial blocks, threshold calibration on only two scenes, incomplete event metadata for size analysis outside Tromsø, and the non-equivariant directional augmentation inherited by the released checkpoints. Sentinel-1 C-band also has an irreducible shadow boundary in very steep relief. Future validation in the Indian Himalaya would be valuable, but this study contains no Indian test data and makes no India-specific accuracy claim. Application to RISAT or NISAR would require sensor-specific calibration and independent Himalayan labels.

6 Conclusion

Terrain-conditioned cross-attention provides competitive Sentinel-1 avalanche-debris segmentation and useful failure diagnostics, but it is not uniformly superior to standard change networks. TopoRadar-Net reaches 63.79% pooled macro F1 across two unseen mountain systems; only its margin over SiamUNet-diff is significant across seeds. Aspect and cross-attention ablations improve pooled F1, and the slope prior suppresses genuine positives. The corridor proxy succeeds in Pish but fails on the only Tromsø corridor event. The defensible conclusion is therefore bounded: radar-topographic conditioning can alter regional precision and recall, while cross-region reliability still requires explicit validation.

Data Availability/Supplementary Information

The AvalCD benchmark is publicly available at <https://doi.org/10.5281/zenodo.15863589>. **Online Resource 1.** Extended methods, complete statistical and subgroup results, and reproducibility identifiers. Code and checkpoint files are publicly archived; the identifying repository URL is withheld from the anonymous manuscript and supplied separately to the editorial office for double-blind review.

507 Statements and Declarations

508

509 **Funding.** No external grant funding was received for this study.

510

511 **Competing interests.** The authors declare no competing financial or non-
512 financial interests.

513

514 **Ethics approval and consent.** Not applicable; the study uses public remote-
515 sensing data and contains no human participants or animals.

516

517 **Consent for publication.** Not applicable.

518

519 **Author contributions.** The complete CRediT statement is provided on the
520 separate title page.

521

522 **AI disclosure.** Generative AI assistance is documented in the Methods section.
523 The authors retain responsibility for all code, analyses, citations, and text.

524

525

526

527

528

529

530 Ahmad, I., Mallick, J., Hang, H.T., Rihan, M., Bakshi, A. (2026). An explainable
531 bayesian-optimized hybrid deep learning framework with attention mechanisms
532 for modeling snow avalanche susceptibility. *Journal of the Indian Society of*
533 *Remote Sensing*, 54(7), 2749–2773, <https://doi.org/10.1007/s12524-026-02459>
534 -1

535

536

537

538

539

540

541 Arjasakusuma, S., & Kusuma, S.S. (2025). Assessment of deep learning based image
542 segmentation for identifying floating net cages from very high-resolution capella
543 synthetic aperture radar (SAR) data. *Journal of the Indian Society of Remote*
544 *Sensing*, 53(2), 485–499, <https://doi.org/10.1007/s12524-024-02022-w>

545

546

547

548

549

550 Bayat, Z., Dindar, S., Erkal, B.G. (2026). Deep learning for post-earthquake road
551 blockage and debris detection in resource-limited contexts. *Journal of the Indian*

552

<i>Society of Remote Sensing</i> , 54(3), 1257–1285, https://doi.org/10.1007/s12524-025-02394-7	553 554 555 556 557 558
Bianchi, F.M., Grahm, J., Eckerstorfer, M., Malnes, E., Vickers, H. (2021). Snow avalanche segmentation in SAR images with fully convolutional neural networks. <i>IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing</i> , 14, 75–82, https://doi.org/10.1109/JSTARS.2020.3036914	559 560 561 562 563 564 565 566 567
Caye Daudt, R., Le Saux, B., Boulch, A. (2018). Fully convolutional siamese networks for change detection. <i>2018 25th IEEE International Conference on Image Processing (ICIP)</i> (pp. 4063–4067).	568 569 570 571 572 573
Chandra, N., Vaidya, H., Chauhan, P., Shaw, R. (2025). <i>YOLOv5 based on attention mechanism for landslide information extraction from multi-source remote sensing images of diverse geological environments</i> . Advance online publication in the Journal of the Indian Society of Remote Sensing.	574 575 576 577 578 579 580
Eckerstorfer, M., Malnes, E., Müller, K. (2017). A complete snow avalanche activity record from a norwegian forecasting region using Sentinel-1 satellite-radar data. <i>Cold Regions Science and Technology</i> , 144, 39–51, https://doi.org/10.1016/j.coldregions.2017.08.004	581 582 583 584 585 586 587 588 589
Eckerstorfer, M., Müller, K., Malnes, E., Oterhals, H.D. (2021). Manual and automatic detection of dry snow avalanches in Sentinel-1 SAR images. <i>Ieee international geoscience and remote sensing symposium (igarss)</i> (pp. 247–250).	590 591 592 593 594 595 596 597 598

599 Eckerstorfer, M., Vickers, H., Malnes, E., Grahn, J. (2019). Near-real time auto-
 600 matic snow avalanche activity monitoring system using Sentinel-1 SAR data in
 601 Norway. *Remote Sensing*, 11(23), 2863, <https://doi.org/10.3390/rs11232863>
 602
 603
 604
 605
 606 European Avalanche Warning Services (2022). *Avalanche size classification standards*
 607 *(classes D1 through D5)*. [https://www.avalanches.org/standards/avalanche](https://www.avalanches.org/standards/avalanche-size/)
 608 [-size/](https://www.avalanches.org/standards/avalanche-size/). (Accessed: September 2026)
 609
 610
 611
 612 Gatti, M. (2026). *AvalCD*. Zenodo.
 613
 614
 615 Gatti, M., Mariani, A., Gallo, I., Monti, F. (2026). *Large-scale avalanche mapping*
 616 *from SAR images with deep learning-based change detection*. arXiv preprint
 617 arXiv:2603.22658.
 618
 619
 620 Gelato, R., Sgaravatti, C., Grahn, J., Boracchi, G., Bianchi, F.M. (2026). Promptable
 621 foundation models for SAR remote sensing: Adapting the segment anything
 622 model for snow avalanche segmentation. *Remote Sensing*, 18(3), 519, [https://](https://doi.org/10.3390/rs18030519)
 623 doi.org/10.3390/rs18030519
 624
 625
 626
 627
 628
 629
 630 Nguyen, H., Dang, K.B., Giang, T.L., Dang, V.B., Dang, N.V., Dao, M.D. (2025).
 631 *Exploring various deep learning models for high-precision landslide tracing in*
 632 *very-high resolution remote sensing imagery*. Advance online publication in the
 633 Journal of the Indian Society of Remote Sensing.
 634
 635
 636
 637 OpenStreetMap contributors (2026). *OpenStreetMap*. [https://www.openstreetmap](https://www.openstreetmap.org/copyright)
 638 [.org/copyright](https://www.openstreetmap.org/copyright). (Major-road snapshot acquired through the Overpass API on
 639 19 September 2026 under ODbL 1.0)
 640
 641
 642
 643
 644

Tompkin, C., & Leinss, S. (2021). Backscatter characteristics of snow avalanches
for mapping with local resolution weighting. *IEEE Journal of Selected Topics
in Applied Earth Observations and Remote Sensing*, 14, 4452–4464, [https://
doi.org/10.1109/JSTARS.2021.3074418](https://doi.org/10.1109/JSTARS.2021.3074418)

Wang, S., Wang, D., Shi, J., Zhang, Z., Wang, H., Li, X., Guo, Y. (2025). Unsupervised
SAR image change detection method based on fused difference images. *Journal
of the Indian Society of Remote Sensing*, 53(12), 4373–4382, [https://doi.org/
10.1007/s12524-025-02259-z](https://doi.org/10.1007/s12524-025-02259-z)

Zhang, Z., Liu, Q., Wang, Y. (2018). Road extraction by deep residual U-Net.
IEEE Geoscience and Remote Sensing Letters, 15(5), 749–753, [https://doi.org/
10.1109/LGRS.2018.2802944](https://doi.org/10.1109/LGRS.2018.2802944)